Consider the following image:

I

0	0	1	4	9
1	0	5	7	11
1	4	9	12	16
3	8	11	14	16
8	10	15	16	20

d/dx

-1	0	1

d/dy

-1
0
1

Compute the Harris matrix

$$H = \sum_{(x,y)\in W} \begin{bmatrix} I_x(x,y)^2 & I_x(x,y)I_y(x,y) \\ I_x(x,y)I_y(x,y) & I_y(x,y)^2 \end{bmatrix}$$

for the 3 by 3 highlighted window. In the above formula $I_x = dI/dx$, $I_y = dI/dy$, and W is the window highlighted in the image.

A) First, compute the derivatives using the differentiation kernels shown above. No normalization (division by 2) is needed.

$I_x = dI/dx$

X	X	X	X	X
X	4	7	6	X
X	8	8	7	X
X	8	6	5	X
X	X	X	X	X

$I_y = dI/dy$

X	X	X	X	X
X	4	8	8	X
X	8	6	7	X
X	6	6	4	X
X	X	X	X	X

B) Now compute the Harris Matrix based on the derivative matrices.

$$H = \sum_{(x,y)\in W} \begin{bmatrix} I_x(x,y)^2 & I_x(x,y)I_y(x,y) \\ I_x(x,y)I_y(x,y) & I_y(x,y)^2 \end{bmatrix}$$

$$\sum_{(x,y)\in W} I_x(x,y)^2 = 4^2 + 7^2 + 6^2 + 8^2 + 8^2 + 7^2 + 8^2 + 6^2 + 5^2 = 403$$

$$\sum_{(x,y)\in W} I_y(x,y)^2 = 4^2 + 8^2 + 8^2 + 8^2 + 6^2 + 7^2 + 6^2 + 6^2 + 4^2 = 381$$

$$\sum_{(x,y)\in W} I_x(x,y)\, I_y(x,y) = 4 * 4 + 7 * 8 + 6 * 8 + 8 * 8 + 8 * 6 + 7 * 7 + 8 * 6 + 6 * 6 + 5 * 4 = 385$$

$$H = \begin{bmatrix} 403 & 385 \\ 385 & 381 \end{bmatrix}$$

C) Compute the Harris cornerness score $C = det(H) - k\ trace(H)^2$ for $k = 0.04$.

Do we have a corner?

$$C = det(H) - K\ trace(H)^2 = 5318 - 0.04 * (784)^2 = -19268.24$$

C is negative with a large magnitude,
such that it is **not** a corner

The matrices in the left column are the output of applying Gaussian filters with different standard derivations for a single octave in the SIFT detection algorithm. There is **a single keypoint.** Your job is to find it based on extrema in scale-space. (the point in the middle scale is considered)

To find the keypoint, you first need to **build the Difference of Gaussian Images** in the scale-space. The key points are found at the locations of **extrema** in scale-space.

Gaussian Filtered Images

25	20	20	16
25	30	19	16
19	17	19	16
13	13	14	14

24	18	20	14
25	32	19	15
18	16	19	14
12	12	13	13

22	15	20	12
24	35	18	14
16	16	18	13
11	11	11	12

20	10	20	8
20	37	10	8
16	8	20	10
12	10	9	10

Gaussian Filtered Images

25	20	20	16
25	30	19	16
19	17	19	16
13	13	14	14

24	18	20	14
25	32	19	15
18	16	19	14
12	12	13	13

22	15	20	12
24	35	18	14
16	16	18	13
11	11	11	12

20	10	20	8
20	37	10	8
16	8	20	10
12	10	9	10

Difference of Gaussian Images

1	2	0	2
0	-2	0	1
1	1	0	2
1	1	1	1

scale = 1

2	3	0	2
1	-3	1	1
2	0	1	1
1	1	2	1

scale = 2

2	5	0	4
4	-2	8	6
0	8	-2	3
-1	1	2	2

scale = 3

The keypoint is located at x=1, y=1, scale=2, Since

Compared to neighbors at scale 1
-3 < 1,2,0,0,-2,0,1,1,0

Compared to neighbors in the same scale (scale 2)
-3 < 2,3,0,1, 1, 2,0,1

Compared to neighbors at scale 3
-3 < 2,5,0,4,-2,8,0,8,-2